

DATA COLLECTION & UNDERSTANDING

1. Data Inventory:

File Name	Purpose	Records	Owner	Refresh Frequency
Orders	Stores customer order transactions, product details, sales, profit, quantity, and discount information.	6000	Business	One-Time Project Dataset
Returns	Stores returned order details and return reasons.	500	Business	One-Time Project Dataset

2. Data Dictionary:

Column Name	Description	Data Type	Example	Nullable
Order ID	Unique identifier for each order	Text	ORD100001	No
Order Date	Date when the order was placed	Date	03-08-2024	No
Ship Date	Date when the order was shipped	Date	10-08-2024	No
Customer ID	Unique identifier for each customer	Text	C2172	No
Customer Name	Name of the customer	Text	Customer 1122	Yes
State	Customer state	Text	Delhi	Yes
Category	Product category	Text	Electronics	Yes
Product ID	Unique identifier for the product	Text	P345	No
Quantity	Number of products ordered	Whole Number	3	No
Unit Price	Price of one unit	Decimal Number	1322	No
Discount	Discount applied to the order	Decimal Number	0.20	Yes

Column Name	Description	Data Type	Example	Nullable
Sales	Total sales amount	Decimal Number	3172.80	No
Profit	Profit earned	Decimal Number	650.40	No

3. Excel Data Quality Checklist:

Check	Status	Observation
Missing Values	Completed	Missing values identified and removed from mandatory fields.
Duplicate Records	Completed	Duplicate records identified and removed.
Date Validation	Completed	Order Date and Ship Date converted to Date format.
Data Types	Completed	Correct data types assigned to all required columns.
Invalid Values	Completed	Invalid Quantity and Discount values corrected.
Text Validation	Completed	Trim and Clean applied to text columns where required.
Blank Records	Completed	Blank/null records removed according to project requirements.
Business Rule Validation	Completed	Quantity values corrected and validated before analysis.